

Math 313 Sp26 007 Linear Algebra

**TuTh 2:00pm – 3:15pm, C E Chavez Bldg. Rm 316**

Instructor Yi Hu Office Hours: Tu 11:00-2:00  
or by appointment

Office: S333, ENR2

Phone: 520 621-4397

E-mail: yhu@arizona.edu

**Text:** Linear Algebra, D Lay S Lay J MacDonald, 6<sup>th</sup> Edi

Homework will be assigned and collected regularly. Your course grade will be calculated based on homework, two midterms, and the final exam.

Lecture 1 Systems of Linear Equations (1.1) + Row Reduction and Echelon Forms (1.2)

Lecture 2 Row Reduction and Echelon Forms (1.2)

Lecture 3 Row Reduction and Echelon Forms (1.2) + Vector Equations, Linear Combinations (1.3)

Lecture 4 The Matrix Equation  $Ax=b$  (1.4)

Lecture 5 Solution Sets of Linear Systems (1.5)

Lecture 6 Linear Independence (1.7)

Lecture 7 Introduction to Linear Transformations (1.8)

Lecture 8 The Matrix of a Linear Transformation (1.9) + Matrix Operations (2.1)

Lecture 9 Matrix Operations (2.1) + The Inverse of a Matrix (2.2)

Lecture 10 The Inverse of a Matrix (2.2) + The Invertible Matrices Theorem (2.3)

Lecture 11 Exam 1 Review / Applications

Exam 1

Lecture 12 Subspaces of  $R^n$  (2.8) + Null Spaces, Column Spaces, Row Spaces (2.8, 4.2)

Lecture 13 Null Spaces, Column Spaces, Row Spaces (2.8, 4.2) + Bases (2.8, 4.3)

Lecture 14 Coordinate Systems (2.9, 4.4) + Dimension and Rank (2.9, 4.5)

Lecture 15 Determinants (3.1, 3.2)

Lecture 16 Determinants (3.1, 3.2) + Determinants and Volume (3.3)

Lecture 17 Exam 2 Review / Applications

Exam 2

Lecture 18 Eigenvectors and Eigenvalues (5.1) + The Characteristic Equation (5.2)

Lecture 19 Diagonalization (5.3)

Lecture 20 Complex Eigenvalues (5.5, App B)

Lecture 21 Inner Product, Length, and Orthogonal sets (6.1, 6.2)

Lecture 22 Orthogonal Projections (6.3)

Lecture 23 Orthogonal Projections (6.3) + The Gram-Schmidt Process (6.4)

Lecture 24 Least Squares Problems (6.5)

Lecture 25 Linear Regression (6.6)

Lecture 26 Final Exam Review / Applications

Lecture 27 Final Exam Review / Applications

Lecture 28 Final Exam Review / Applications

Final Exam